10 INHERITANCE • DETAILED SOLUTIONS

output solutions

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

Detailed Solutions — Predict the Output & Debugging (Module 10: Inheritance)

Solution to Q7

- `d.show()` calls `derived::show()` (`Derived`).
- `d.base::show()` explicitly calls `base::show()` via the scope resolution operator (`Base`).
- **Guaranteed Output**:

```
TEXT
Derived Base
```

Solution to Q8

- In `class C : public B, public A` , `B` is listed before `A` in the class header.
- Therefore, constructors run `B()` , then `A()` , then `C()` .
- Destructors run in reverse: `~C()` , then `~A()` , then `~B()` .
- **Guaranteed Output**:

```
TEXT
B A C ~C ~A ~B
```

Solution to Q9

- **Rule (`[class.base.init]`)**: Virtual base classes are always initialized **first**, before any non-virtual base classes!
- Step 1: `B` has a virtual base `V` , so the shared virtual base `V()` is constructed first (`V`).
- Step 2: Non-virtual bases of `D` are initialized in declaration order (`A` , then `B`).
  - `A` inherits `V` *non-virtually*, so `A` 's own separate `V` subobject is constructed (`V`), followed by `A()` (`A`).
  - `B` 's virtual base `V` was already constructed in Step 1, so `B()` runs (`B`).
- Step 3: `D()` runs (`D`).
- **Guaranteed Output**:

```
TEXT
V V A B D
```

Solution to Q10 (Debugging `private` Base Member)

- **Error**: `age` is `private` in `person`, so `student::set(int)` cannot access `age` directly even with `public` inheritance.
- **Fix**: Change `private:` to `protected:` in `class person` (or provide a `public: void set_age(int)` member function in `person`).

Solution to Q11 (Debugging Multiple Inheritance Ambiguity)

- **Error**: Both `father` and `mother` define `void greet()`, so `obj.greet()` is ambiguous.
- **Fix**: Disambiguate using the scope resolution operator in `main()`:

```cpp
obj.father::greet();
obj.mother::greet();
```